

Tracking and Positioning Dynamic Targets in VLC: Signal Dependent Noises and Bayesian Bound Analysis

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Abstract—This paper investigates the problem of position estimation and tracking in indoor visible light communication (VLC) systems under a multiple-input single-output (MISO) configuration. The study specifically addresses the impact of signal-dependent shot noise (SDSN) and relative intensity noise (RIN), which significantly degrade localization accuracy in practical VLC scenarios. To establish a theoretical performance benchmark, a novel closed-form Bayesian Cramér-Rao lower bound (BCRLB) is derived for two-dimensional (2D) position estimation. The proposed extended Kalman filter (EKF) demonstrates superior localization accuracy by effectively accounting for motion dynamics through its recursive updates, unlike estimation methods that rely solely on noisy distance measurements. Our findings show that increasing the number of light sources improves spatial diversity, thereby enhancing positioning precision. Furthermore, analysis indicates that RIN has a more pronounced negative effect on tracking accuracy than SDSN. The EKF's performance is validated against the proposed BCRLB, confirming its suitability for real-time and reliable indoor localization. Overall, this work provides meaningful insights into the design of robust and efficient VLC-based positioning systems.

Index Terms—Visible light communications, tracking, Kalman, Fisher information matrix, Bayesian, Cramér-Rao bound, relative intensity noise, signal-dependent shot noise.

I. INTRODUCTION

THE rapidly increasing number of localization services provides a wide range of applications across business, industry, and everyday life. To have an accurate, rapid, and reliable localization is a critical problem in many applications such as wireless sensor networks, robotics, and navigation [1].

A positioning system allows a mobile target to identify its location and provides this information for various position-based services. Indoor localization applications eliminate the

use of the global positioning system (GPS) because it cannot penetrate buildings, which means no indoor coverage. In addition, GPS offers only meter-range accuracy, while these applications demand a much higher level of precision, often requiring accuracy within the centimeter range. As a result, efforts are directed towards finding effective indoor localization techniques. These techniques leverage a range of technologies [2].

Specifically, ultra-wideband (UWB), radio-frequency identification (RFID), infrared (IR), and ultrasound are proposed to be used in indoor localization. In these technologies, the installation of Wi-Fi access points (APs) or infrared transmitter units is often required, resulting in considerable expenses for setup. In addition, these radio frequency (RF) based techniques can not be used in some RF interference-sensitive locations, such as aircraft cabins, hospitals, and petrochemical factories. Moreover, multipath interference can greatly impact the performance of RF-based positioning, with its effects being particularly pronounced in indoor positioning [3].

In contrast to traditional indoor wireless positioning systems, visible light positioning (VLP), which uses visible light communication (VLC) to determine the accurate location of a user, provides a wide range of unlicensed bandwidth in Terahertz (THz), enabling high levels of localization accuracy. In addition, unlike the RF signal, the optical signal is less susceptible to multipath interference, making VLP potentially more accurate than RF-based positioning. Moreover, VLC bandwidth from 430 THz to 790 THz does not interfere with the RF spectrum from 3 KHz to 3 GHz or mm-wave from 30 GHz to 60 GHz, which makes it suitable for RF/mm wave-restricted places. VLP uses visible light from existing lighting infrastructure for positioning, which means the light is utilized for both communication and illumination. This dual-purpose use makes VLP an energy-efficient and cost-effective position technology [4]. Despite the numerous advantages of VLP, there are additional sources of noise unique to VLP systems that do not exist in RF positioning systems. These include signal-dependent shot noise (SDSN) and relative intensity noise (RIN), all of which can significantly degrade the accuracy of positioning. These unique noise sources create challenges for VLP systems, requiring careful system design to ensure reliable positioning performance [5].

One of the most distinguished differences between RF and VLC is the existence of SDSN in VLC systems. SDSN is

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a type of shot noise whose intensity is directly related to the signal strength. It originates from the random arrival of photons at the receiver in optical systems, such as VLC [6], [7]. In such systems, the number of photons that arrive at the photodetector (PD) varies randomly due to the quantum nature of the received light. This unique characteristic of SDSN makes it distinct from other types of noise, such as thermal noise, which remains relatively constant regardless of signal strength [8], [9]. Although SDSN originally follows the Poisson distribution due to the stochastic nature of photon arrivals, it is often modeled as a Gaussian-distributed random variable in VLC systems. This can be explained by the large number of photons typically present in VLC signals, allowing the Poisson distribution to be fairly approximated to a Gaussian one [10].

RIN is another significant factor that distinguishes VLC from RF communication. RIN arises from fluctuations in the intensity of light emitted by the light sources used in VLC systems. Its variance is directly proportional to the square of the transmission signal power. These variations, often due to the inherent properties of the light source or environmental factors, introduce additional noise into the communication channel [11], [12]. In [13], RIN was experimentally measured in semiconductor lasers and compared to the noise characteristics observed in light-emitting diodes (LEDs) and superluminescent diodes (SLDs). SLDs are optical sources that combine structural features of both LEDs and laser diodes. As a result, they exhibit both RIN and SDSN due to their internal optical gain without optical feedback, as discussed in [14, Chapter 2], [15], and [16]. In VLP systems, various types of transmitters, such as fluorescent lamps, LEDs, micro-LEDs, and laser diodes, are employed, each offering unique emission properties, modulation capabilities, and noise characteristics that impact system design and positioning performance [17].

A. Literature Review

Interest in VLP research has recently increased because of advances in the light sources industry. The lighting industry has been revolutionized by the widespread adoption of more economical and efficient light sources. These light sources offer energy savings and longer lifespans and provide the high-frequency modulation capabilities necessary for precise VLP systems. This technological evolution has opened up new possibilities for indoor localization-based applications, making VLP a highly attractive area for further research and development. In VLP systems, several algorithms have been developed, each with distinct advantages and limitations. Proximity-based methods are simple and low-cost but offer limited accuracy. Time of arrival (TOA), time difference of arrival (TDOA), and angle of arrival (AOA) techniques provide higher accuracy but require multiple LEDs, time synchronization, and complex hardware. Fingerprinting and image-sensing methods deliver high accuracy with fewer LEDs but suffer from poor portability and slower response. Hybrid methods improve robustness by combining techniques, but increase system complexity. In this work, we adopt the received signal strength (RSS)-based approach due to its simplicity and low cost. However, since it relies on the channel model, we

implement recursive estimation techniques to achieve robust positioning accuracy in localization and tracking [8], [18].

The authors in [19] presented and validated an AOA-based VLP system using a specific type of receiver, a quadrant solar cell (QSC). To enhance localization accuracy, they integrated third-order ridge regression machine learning (RRML) with the QSC. Experimental results demonstrated that the root-mean-square (RMS) of the average position error was reduced from 7.2177 cm to 3.2025 cm using third-order regression and further improved to 3.0881 cm with the proposed third-order RRML. This approach highlights the potential of combining AOA techniques with machine learning; however, it also relies on specialized hardware, which may limit system generalizability and increase implementation cost.

While in [20], the AOA estimators with PDs were used to accurately determine beacon LED positions in VLP systems, improving localization accuracy by mitigating errors from imprecise LED placements. AOA algorithms determine the receiver's position by utilizing the estimated signals' arrival angles from multiple light sources. Consequently, these algorithms are susceptible to the accuracy of the estimated angles, which can greatly influence their overall accuracy. Authors in [21] present a VLP system combining a passive time of flight (ToF) camera with field-programmable gate array (FPGA)-modulated LEDs, using AOA and TDOA algorithms for accurate indoor localization. In [22], a TOA-based system is proposed for smartphone positioning using modulated LEDs and acoustic signals. Both studies focus on experimental setups and do not provide analytical performance models or consider SDSN or RIN. Similarly, [23] applied a TOA scheme for user localization and synchronization in a periodic asymmetric ranging network, where a primary anchor node transmitted signals and secondary nodes received them.

To address the limitations and complexity of AOA and TOA techniques, the RSS method has been proposed and is widely utilized. This approach estimates the user's location based on the power of the received signal, as RSS is considered a straightforward and efficient localization technique. In [24], authors used a VLP implementation to estimate the position of a vehicle in a room using the RSS scheme and fixed LED-based light transmitters. In [25], the authors proposed a camera-assisted RSS algorithm for VLP systems to improve the accuracy of the positioning of the user. Moreover, in [26], an iterative least squares algorithm is introduced for static indoor VLP based on RSS, with consideration of SDSN. Nonetheless, the framework does not include the impact of RIN, nor does it address dynamic VLP scenarios involving time-varying receiver positions. Although many studies in the literature have studied the localization of a user using VLP, no work has considered studying the effect of the joint presence of SDSN and RIN.

Recently, a number of studies have applied the extended Kalman filter (EKF) to enhance VLP. For instance, [27] proposes an EKF-based solution for PD-based VLP, where the measurement model incorporates Lambertian emission characteristics and geometric constraints. Similarly, [28] improves RSS-based localization by combining temporal averaging with Kalman filtering to reduce measurement variability. In [29], a

fusion framework integrates VLP measurements with inertial navigation using EKF to maintain continuous tracking even in the presence of LED signal blockage and multipath effects. More recently, [30] introduces a joint position and orientation estimation scheme that fuses angle and distance information from multiple LEDs within an EKF framework, achieving improved accuracy in dynamic environments. Despite their contributions, these studies do not consider the effect of signal-dependent noise, such as SDSN and RIN, which are inherent in optical wireless systems and can substantially impact tracking performance. Furthermore, none of these works provides a mathematical benchmark like the Bayesian Cramér-Rao Lower Bound (BCRLB), which is essential for evaluating the theoretical accuracy limits of estimators, especially in recursive systems such as those relying on EKF.

A rigorous theoretical framework, such as the Cramér-Rao lower bound (CRLB), is essential for analyzing and optimizing tracking performance in VLP systems. This framework provides a benchmark to assess the limits of accuracy and guide the development of more effective positioning techniques. A closed-form CRLB for the user location and orientation of non-line-of-sight (NLoS) propagation on the RSS-based VLP was derived in [31]. While in [2], the CRLB of the positioning error of a hybrid RF-VLC and positioning system using the RSS measurement method for positioning was derived. In [32], the authors derived the CRLB for the integrated visible light positioning and communication (VLPC) system using an RSS-based 3D positioning scheme. To the best of the authors' knowledge, there is no work in the literature that considers the derivation of the BCRLB for location error in the presence of SDSN and RIN in a VLP system.

B. Key Contributions

The key contributions of this paper can be summarized as follows:

- 1) *First Investigation of SDSN and RIN in VLC Localization*: This study is the first to systematically investigate the impact of SDSN and RIN on position estimation and tracking in VLC systems. We provide a comprehensive analysis highlighting the distinct characteristics and combined effects of these noise types.
- 2) *Closed-form BCRLB Derivation*: We derive, for the first time, a closed-form BCRLB for VLP systems that accounts for both SDSN and RIN. This theoretical benchmark offers new insights into the fundamental limits of localization accuracy under realistic VLC noise conditions.
- 3) *EKF-Based Tracking Under Combined Noise*: We implement the EKF to track a moving PD target. To the best of our knowledge, this is the first work to apply EKF in the presence of both SDSN and RIN. The EKF effectively combines prediction and noisy measurement updates, significantly improving robustness over conventional estimation methods.
- 4) *Impact of light source density in multiple-input single-output (MISO) VLC systems*: We investigate how increasing the number of light sources in MISO VLC

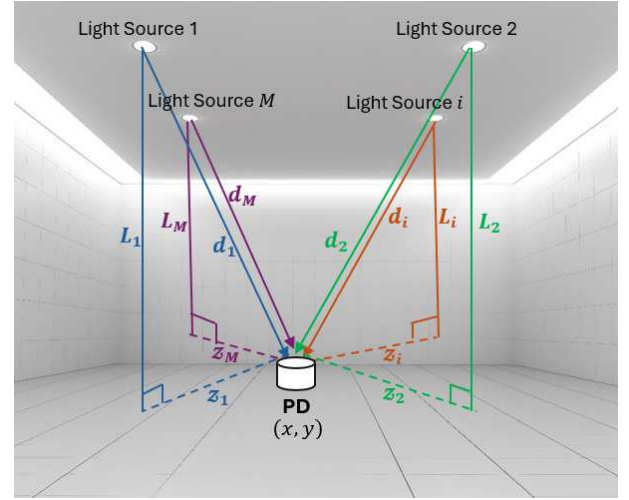


Fig. 1. Visible light positioning system model. The system consists of M light sources and a single photodetector (PD). The light sources are used to estimate the position of the PD based on the received signal strength.

systems affects localization performance. Our results demonstrate that a higher light source density enhances measurement diversity and improves estimation accuracy, especially when the target lies within overlapping light source coverage areas.

- 5) *Comparison with Benchmark Estimation Techniques*: We compare the performance of our proposed position tracking and estimation approach with established techniques such as recursive Levenberg–Marquardt (RLM) and particle filtering (PF). The results show that EKF outperforms RLM in terms of position error bounds and estimation reliability. While PF achieves performance close to EKF, the EKF offers significantly lower computational complexity and is better suited for Gaussian noise models like ours.

C. Organization

The rest of the paper is organized as follows: The system model is overviewed in II. In III, the BCLRB derivation is introduced. Next, the various estimation and tracking methods of the position of a moving target used in this paper are explained in IV. The simulation results are presented and discussed in V, and finally, this work is concluded in VI.

II. SYSTEM AND CHANNEL MODEL

We consider an indoor MISO VLP system consisting of M light sources fixed on a room's ceiling, serving a single user equipped with a PD, as shown in Fig. 1. We assume that the user moves with a known velocity $\mathbf{v}$. Each light source transmits a visible light signal to the PD. The transmitted signal from each source is denoted as x , and, without loss of generality, we assume that all sources transmit with the same power, thus $x = p > 0$ [33]. The received signal vector $\mathbf{r} = [r_1, r_2, \dots, r_i, \dots, r_M]^T$ can be expressed as:

$$\mathbf{r} = \underbrace{\mathbf{h}\mathbf{x}}_{\text{Signal}} + \underbrace{\sqrt{x \text{diag}(\mathbf{h})} \mathbf{n}_{\text{ds}}}_{\text{SDSN}} + \underbrace{x \text{diag}(\mathbf{h}) \mathbf{n}_R}_{\text{RIN}} + \underbrace{\mathbf{n}}_{\text{Thermal Noise}}, \quad (1)$$

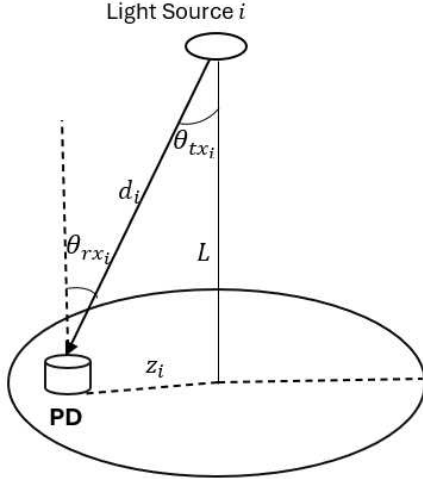


Fig. 2. Visible light communication channel model.

where M is the number of transmitted light sources, $\text{diag}(\mathbf{h})$ is an $M \times M$ diagonal matrix with the elements of the channel gain vector $\mathbf{h} = [h_1, h_2, \dots, h_i, \dots, h_M]^T$ on the main diagonal. The noise terms are as follows: $\mathbf{n} = [n_1, n_2, \dots, n_i, \dots, n_M]^T$, is the thermal noise and SDSN vector is $\mathbf{n}_{ds} = [n_{ds1}, n_{ds2}, \dots, n_{dsi}, \dots, n_{dsM}]^T$ and finally RIN $\mathbf{n}_R = [n_{R1}, n_{R2}, \dots, n_{Ri}, \dots, n_{RM}]^T$ [34], [35].

The elements of $\mathbf{n}$ are independent and identically distributed Gaussian random variables, i.e., $\mathbf{n} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I}_M)$. Similarly, we have $\mathbf{n}_{ds} \sim \mathcal{N}(0, \sigma_{ds}^2 \mathbf{I}_M)$ and $\mathbf{n}_R \sim \mathcal{N}(0, \sigma_R^2 \mathbf{I}_M)$. The channel gain h_i between the i th light source and the user can be expressed as [11]:

$$h_i = \frac{A_{pd_i} \eta_i (m_i + 1)}{2\pi d_i^2} \cos^m(\phi_{tx_i}) T_s(\theta_{rx_i}) g(\theta_{rx_i}) \cos(\theta_{rx_i}), \quad (2)$$

where d_i is the Euclidean distance from the transmitter light source to the receiver, which can be calculated as $d_i = \sqrt{L^2 + z_i^2}$, where L corresponds to the vertical distance from the i^{th} light source to the PD surface, and z_i is the horizontal distance from the center of the light source cell to the PD, as illustrated in Fig. 2, and ϕ_{tx_i} represents the angle of irradiance. It's worth mentioning that $\cos(\phi_{tx_i}) = \cos(\theta_{rx_i}) = \frac{L}{d_i}$, in the case when the receiver is oriented upwards and the transmitter is directed downwards, θ_{rx_i} signifies the angle of incidence surface, $T_s(\theta_{rx_i})$ stands for the gain associated with the receiver's optical filter, $g(\theta_{rx_i}) = n_i^2 / \sin^2(\phi_{FOV_i})$ describes the gain of the optical concentrator, which relies on the refractive index n_i of the concentrator and the field-of-view (FOV) angle, i.e., ϕ_{FOV_i} . It's important to note that $g(\theta_{rx_i}) = 0$ when $\theta_{rx_i} \geq \phi_{FOV_i}$ [6], [36]. Additionally, A_{pd_i} denotes the detection area of the PD and η represents the average responsivity of the receiver and finally $m_i = \frac{-1}{\log_2(\cos(\Phi_{i1/2}))}$ is the Lambertian radiation order and is determined by the light source emission semi-angle $\Phi_{i1/2}$.

Hence, the received signal vector can be rewritten as

$$\mathbf{r} = (\mathbf{c} \odot \boldsymbol{\beta})p + \sqrt{\text{diag}(\mathbf{c} \odot \boldsymbol{\beta})} \mathbf{p} \mathbf{n}_{ds} + \text{diag}(\mathbf{c} \odot \boldsymbol{\beta}) \mathbf{p} \mathbf{n}_R + \mathbf{n}, \quad (3)$$

where $\boldsymbol{\beta} = [d_1^{-\alpha}, d_2^{-\alpha}, \dots, d_i^{-\alpha}, \dots, d_M^{-\alpha}]^T$, $\odot$ is the Hadamard product operator. The constant vector $\mathbf{c} = [c_1, c_2, \dots, c_i, \dots, c_M]^T$ can be calculated as

$$c_i = \frac{A_{pd_i} \eta_i (m_i + 1)}{2\pi} L^{m_i+1} T_s(\theta_{rx_i}) g(\theta_{rx_i}), \quad (4)$$

where $\alpha = m + 3$. It's worth noting that the distortion variance is directly influenced by the distance between the transmitter and the receiver. This sets the additive distortion term apart from conventional receiver noise, which remains independent of distance. It is worth mentioning that in the ideal case, the received signal can be written as $\mathbf{r} = (\mathbf{c} \odot \boldsymbol{\beta})p + \mathbf{n}$.

III. PD TRANSITION MODEL AND BAYESIAN CRAMÉR-RAO LOWER BOUND

In this section, we consider a PD moving in a 2D x - y plane. Since the PD's position changes over time, all measurements become time-dependent. We also present a closed-form expression for the BCRLB, which serves as a fundamental performance benchmark for position estimation under dynamic conditions. Because the estimation depends on the PD's motion over time, the BCRLB must be updated recursively as the PD's position changes. To derive the BCRLB, we first determine the Fisher information matrix (FIM) of the estimated distances, referred to as the distance Fisher information matrix (DFIM). Using the DFIM, we determined the FIM for the PD's position, denoted as the position Fisher information matrix (PFIM). Finally, with the PFIM in hand and incorporating the PD's dynamic motion, we are ultimately able to compute the time-varying BCRLB. To the best of the authors' knowledge, the previous bounds have not been determined, even for the ideal case, i.e., without SDSN and RIN.

A. PD Transition Model

In this subsection, we outline the mathematical model for the motion of PD within an indoor environment illuminated by multiple light source transmitters. While the PD moves, its position may experience slight deviations from the ideal path due to interactions with environmental obstacles and objects. The PD collects distance measurements from the light source transmitters at uniform intervals of Δ seconds, and these measurements are utilized to model and analyze its dynamic position over time [37], [38].

For the 2D motion model, the position of the PD at the t -th sample can be expressed as: $\mathbf{P}[t] = [x[t], y[t]]^T \in \mathbb{R}^2$. Therefore, the transition model governing the motion of the PD can be written as:

$$\mathbf{P}[t] = \mathbf{P}[t-1] + \mathbf{q}[t] + \mathbf{u}[t], \quad (5)$$

where $\mathbf{q}[t]$ denotes the control input associated with motion under time-varying velocity, defined as: $\mathbf{q}[t] = \Delta \mathbf{v}[t-1] + \frac{1}{2} \mathbf{a} \Delta^2$, with $\mathbf{v}[t] = [v_x[t], v_y[t]]^T$ representing the velocity of the PD in the 2D x - y plane and $\mathbf{a} = [a_x, a_y]^T$ denoting the acceleration [39]. The term $\mathbf{u}[t]$ captures random perturbations in the motion due to unexpected environmental factors.

The process noise $\mathbf{u}[t]$ is modeled as a Gaussian random vector with zero mean and covariance matrix $\mathbf{Q}$, such that:

$$\mathbf{u}[t] \sim \mathcal{N}(0, \mathbf{Q}). \quad (6)$$

For simplicity, assume that $\mathbf{Q} = [\sigma_x^2, \sigma_y^2]^T$, where σ_x^2 and σ_y^2 denote the variances of the noise along the x and y axes, respectively. For any two samples $t \neq t'$, the process noise vectors $\mathbf{u}[t]$ and $\mathbf{u}[t']$ are independent, ensuring that:

$$\mathbb{E}[\mathbf{u}[t] \mathbf{u}^T[t']] = 0 \text{ for } t \neq t'. \quad (7)$$

This independence implies that $\mathbf{u}[t]$ is a vector of additive white Gaussian noise [40], [41]. Because the noise vectors are independent across time, the position sequence $\mathbf{P}[t]$ forms a Markov chain. As a result, this Markov property can be formulated as:

$$p(\mathbf{P}[t] | \mathbf{P}[0:t-1]) = p(\mathbf{P}[t] | \mathbf{P}[t-1]), \quad (8)$$

indicating that the conditional probability of the PD's position at time t depends only on its immediate previous state at time $t-1$. By leveraging the Gaussian nature of $\mathbf{u}[t]$, this conditional probability distribution is given by:

$$p(\mathbf{P}[t] | \mathbf{P}[t-1]) \sim \mathcal{N}(\mathbf{P}[t-1] + \mathbf{q}[t], \mathbf{Q}). \quad (9)$$

In terms of the initial conditions, the initial state of the position $\mathbf{P}[-1]$ is independent of $\mathbf{u}[t]$ and follows a Gaussian distribution:

$$\mathbf{P}[-1] \sim \mathcal{N}(\mathbf{P}_0 + \mathbf{q}_0, \mathbf{Q}_0). \quad (10)$$

Here $\mathbf{P}_0$, $\mathbf{Q}_0$, and $\mathbf{q}_0$ can be selected arbitrarily or derived from prior knowledge. These initial values are then provided as inputs to the EKF as we will describe in the subsequent section. As the number of samples increases, the effect of the initial conditions becomes negligible [42], [43].

B. Distance Fisher Information Matrix (DFIM)

To establish a fundamental benchmark for dynamic position estimation, first, we derive DFIM based on the received signal, denoted as $\mathbf{J}_d^{(O)}[t]$. The DFIM quantifies the relationship between the information contained in the received signal and the distances between the transmitting light source and the PD receiver. The matrix can be expressed as follows:

$$\mathbf{J}_d^{(O)}[t] = -\mathbb{E} \left[\frac{\partial^2}{\partial \mathbf{d}[t] \partial \mathbf{d}[t]^T} \ln f(\mathbf{r}[t] | \mathbf{d}[t]) \right], \quad (11)$$

where $\mathbb{E}[\cdot]$ is the expectation operator and $\mathbf{r}[t]$ is a M -dimensional Gaussian random vector with mean vector $\boldsymbol{\mu}[t] = (\mathbf{c} \odot \boldsymbol{\beta}[t])p$, where $\mathbf{c}$ and $\boldsymbol{\beta}[t]$ are system-specific parameters, and covariance matrix $\boldsymbol{\Sigma}_s[t] = \text{diag}[\sigma_{s1}^2[t], \dots, \sigma_{si}^2[t], \dots, \sigma_{sM}^2[t]]$, where each variance term is $\sigma_{si}^2[t] = \sigma_{ni}^2 + \sigma_{d_{si}}^2 c_i p_i d_i[t]^{-\alpha} + \sigma_{Ri}^2 c_i^2 p_i^2 d_i[t]^{-2\alpha}$, representing the variance of the i -th observation, it follows that: $\mathbf{r}[t] \sim \mathcal{N}(\boldsymbol{\mu}[t], \boldsymbol{\Sigma}_s[t])$.

The probability density function (PDF) of $\mathbf{r}[t]$ conditioned on $\mathbf{d}[t]$ can be depicted as:

$$f(\mathbf{r}[t] | \mathbf{d}[t]) = \frac{1}{(2\pi)^{\frac{M}{2}} |\boldsymbol{\Sigma}_s[t]|^{1/2}} \times$$

$$\exp \left(-\frac{1}{2} (\mathbf{r}[t] - \boldsymbol{\mu}[t])^T \boldsymbol{\Sigma}_s[t]^{-1} (\mathbf{r}[t] - \boldsymbol{\mu}[t]) \right). \quad (12)$$

Therefore, the DFIM ($\mathbf{J}_d^{(O)}[t]$) is an $M \times M$ matrix, and each element in it can be computed as:

$$\begin{aligned} \mathbf{J}_d^{(O)}[t]_{(i,j)} &= \frac{\partial \boldsymbol{\mu}[t]^T}{\partial d_i[t]} \boldsymbol{\Sigma}_s[t]^{-1} \frac{\partial \boldsymbol{\mu}[t]}{\partial d_j[t]} \\ &+ \frac{1}{2} \text{tr} \left(\boldsymbol{\Sigma}_s[t]^{-1} \frac{\partial \boldsymbol{\Sigma}_s[t]}{\partial d_i[t]} \boldsymbol{\Sigma}_s[t]^{-1} \frac{\partial \boldsymbol{\Sigma}_s[t]}{\partial d_j[t]} \right), \end{aligned} \quad (13)$$

where $\text{tr}(\cdot)$ is the trace operator, and the derivatives of the mean and covariance matrix are given as follows:

$$\frac{\partial \boldsymbol{\mu}[t]}{\partial d_i[t]} = [0, \dots, -p\alpha c_i d_i[t]^{-(\alpha+1)}, \dots, 0]^T, \quad (14)$$

$$\begin{aligned} \frac{\partial \boldsymbol{\Sigma}_s[t]}{\partial d_i[t]} &= \text{diag} \left[0, \dots, (-\alpha \sigma_{d_{si}}^2 c_i p_i d_i[t]^{-(\alpha+1)} \right. \\ &\quad \left. - 2\alpha \sigma_{Ri}^2 c_i^2 p_i^2 d_i[t]^{-(2\alpha+1)}), \dots, 0 \right], \end{aligned} \quad (15)$$

furthermore $\boldsymbol{\Sigma}_s[t]^{-1}$ is expressed as:

$$\boldsymbol{\Sigma}_s[t]^{-1} = \text{diag} \left[\frac{1}{\sigma_{s1}^2[t]}, \dots, \frac{1}{\sigma_{si}^2[t]}, \dots, \frac{1}{\sigma_{sM}^2[t]} \right]. \quad (16)$$

Finally, after substituting the derived expressions and the inverse covariance matrix $\boldsymbol{\Sigma}_s[t]^{-1}$ into (13), then carrying out the necessary summations and algebraic manipulations, we obtain the full expression of each element of DFIM ($\mathbf{J}_d^{(O)}[t]_{(i,j)}$). It is worth mentioning that this matrix is diagonal due to the statistical independence between the estimated distances, and its elements can be calculated as:

$$\begin{aligned} \mathbf{J}_d^{(O)}[t]_{(i,i)} &= \frac{p^2 c_i^2 \alpha^2 d_i[t]^{-2(\alpha+1)}}{\sigma_{si}^2[t]} \\ &+ \frac{1}{2} \frac{(-\sigma_{d_{si}}^2 c_i p \alpha d_i[t]^{-(\alpha+1)} - 2\sigma_{Ri}^2 c_i^2 p^2 \alpha d_i[t]^{-(2\alpha+1)})^2}{\sigma_{si}^4[t]}. \end{aligned} \quad (17)$$

The following remarks provide a deeper understanding of how noise and various parameters in (17) influence the calculation of the DFIM.

Remark 1: As the distance $d_i[t]$ increases, the value of $\mathbf{J}_d^{(O)}[t]_{(i,i)}$ decreases, resulting in a reduction of the corresponding diagonal element of the DFIM. Therefore, a reduction in the element of DFIM corresponds to an increase in the variance of the distance estimation error. This means that the accuracy of the estimated distance decreases as the PD moves farther away from the light source.

Remark 2: When $\sigma_{Ri}^2 = 0$ and $\sigma_{d_{si}}^2 = 0$, i.e., ideal case, for all estimated distances, the value of the corresponding diagonal elements of DFIM increases proportionally to the square of the transmitted power, as given by $\mathbf{J}_d^{(O)}[t]_{(i,i)} = \frac{p^2 c_i^2 \alpha^2 d_i[t]^{-2(\alpha+1)}}{\sigma_{ni}^2}$. In this scenario, the dominant noise source is thermal noise, which is independent of the transmitted signal's power. Therefore, increase in transmitted power can lead to higher estimation accuracy.

Remark 3: In (17), the expression for $\mathbf{J}_d^{(O)}[t]_{(i,i)}$ under the condition $\sigma_{Ri}^2 \rightarrow \infty$ simplifies to $\lim_{\sigma_{Ri}^2 \rightarrow \infty} \mathbf{J}_d^{(O)}[t]_{(i,i)} = 2\alpha^2 d_i[t]^{-2}$. This indicates that, in the presence of high RIN, $\mathbf{J}_d^{(O)}[t]_{(i,i)}$ primarily depends on the distance between the transmitter and receiver and the material of the receiver. The influence of transmitted power becomes negligible, leading to a saturation level as the power increases.

Remark 4: In this scenario, taking the limit as power p tends to infinity: $\lim_{p \rightarrow \infty} \mathbf{J}_d^{(O)}[t]_{(i,i)} = \alpha^2 d_i[t]^{-2} \left(\frac{1}{\sigma_{Ri}^2} + 2 \right)$. If this happens to all measured distances in DFIM. This implies that the diagonal elements of DFIM reach a saturation level where its value depends on both σ_{Ri}^2 and the distance. Even with unlimited power, the impact of σ_{Ri}^2 remains significant. As σ_{Ri}^2 grows large, the system settles at this saturation point, rendering any further increase in power inconsequential.

Remark 5: In (17), when $\sigma_{dsi}^2 \rightarrow \infty$, the expression for $\mathbf{J}_d^{(O)}[t]_{(i,i)}$ simplifies to: $\lim_{\sigma_{dsi}^2 \rightarrow \infty} \mathbf{J}_d^{(O)}[t]_{(i,i)} = \frac{1}{2}\alpha^2 d_i[t]^{-2}$. This indicates that in cases dominated by σ_{dsi}^2 if we apply the same condition to all distances, the diagonal elements of DFIM become dependent on the distance and fabrication material of the receiver only.

Remark 6: In (17), when $\sigma_{Ri}^2 \rightarrow 0$, and $p \rightarrow \infty$ the expression for $\mathbf{J}_d^{(O)}[t]_{(i,i)}$ is simplified to $\mathbf{J}_d^{(O)}[t]_{(i,i)} = \frac{pc_i \alpha^2 d_i[t]^{-(\alpha+2)}}{\sigma_{dsi}^2} + \frac{1}{2}\alpha^2 d_i[t]^{-2}$. This result demonstrates that in the absence of σ_{Ri}^2 and as $p \rightarrow \infty$, the value of the corresponding diagonal elements of DFIM increases linearly with p . Additionally, in the same equation when $\sigma_{dsi}^2 \rightarrow 0$, and $p \rightarrow \infty$ the expression for $\mathbf{J}_d^{(O)}[t]_{(i,i)}$ is simplified to $\mathbf{J}_d^{(O)}[t]_{(i,i)} = \alpha^2 d_i[t]^{-2} \left(2 + \frac{1}{\sigma_{Ri}^2} \right)$. This result demonstrates that in the absence of σ_{dsi}^2 and as $p \rightarrow \infty$, the value of the corresponding diagonal elements of DFIM reaches an instant saturation level, and increasing power does not affect them. However, as discussed in Remark 2, when both $\sigma_{Ri}^2 \rightarrow 0$ and $\sigma_{dsi}^2 \rightarrow 0$, the diagonal elements of DFIM increase proportionally with p^2 .

C. Position Fisher Information Matrix (PFIM)

Since our goal is to estimate the position of a dynamic PD we have derived a PFIM. This calculation is carried out by utilizing the DFIM and the known relationship between the measured distances and the x - y coordinates of the target. Specifically, each measured distance is related to the target's position by $d_i[t] = \sqrt{(x[t] - x_i)^2 + (y[t] - y_i)^2}$. The PFIM of the target position, denoted as $\mathbf{J}_P^{(O)}[t]$, has dimensions 2×2 . To bridge the gap between the DFIM and the PFIM, we introduce the Jacobian matrix $\mathbf{T}[t] \in \mathbb{R}^{2 \times M}$, which captures how variations in the target's x - y coordinates affect the measured distances and is defined as:

$$\mathbf{J}_P^{(O)}[t] = \mathbf{T}[t] \mathbf{J}_d^{(O)}[t] \mathbf{T}[t]^T, \quad (18)$$

$$\mathbf{T}[t] = \frac{\partial \mathbf{d}^T[t]}{\partial \mathbf{P}[t]} = \begin{bmatrix} \frac{\partial d_1[t]}{\partial x[t]} & \dots & \frac{\partial d_i[t]}{\partial x[t]} & \dots & \frac{\partial d_M[t]}{\partial x[t]} \\ \frac{\partial d_1[t]}{\partial y[t]} & \dots & \frac{\partial d_i[t]}{\partial y[t]} & \dots & \frac{\partial d_M[t]}{\partial y[t]} \end{bmatrix}, \quad (19)$$

where the partial derivatives with respect to x and y are given by:

$$\frac{\partial d_i[t]}{\partial x[t]} = \frac{(x[t] - x_i)}{\sqrt{(x[t] - x_i)^2 + (y[t] - y_i)^2}}, \quad (20)$$

$$\frac{\partial d_i[t]}{\partial y[t]} = \frac{(y[t] - y_i)}{\sqrt{(x[t] - x_i)^2 + (y[t] - y_i)^2}}. \quad (21)$$

After that substitute values of $\mathbf{T}[t]$, $\mathbf{J}_d^{(O)}[t]$ in (18) the PFIM of location $\mathbf{J}_P^{(O)}[t]$ can be expressed in (22), as shown at the bottom of the page.

Remark 7: PFIM in (22) depends on the number of light sources, M . As the number of light sources increases, more terms are added to the summation, increasing the value of each element in PFIM and, consequently, decreasing the CRLB of the estimated position. This implies that increasing M improves the accuracy of position estimation. It is worth mentioning that the minimum number of light sources is three; otherwise, the PFIM becomes singular, making it impossible to determine the CRLB matrix. In other words, the position estimation cannot be performed.

D. Bayesian Cramér-Rao Lower Bound (BCRLB)

The recursive or BCRLB is a theoretical performance limit on the accuracy of parameter estimates in dynamic, time-varying systems. Unlike the classical CRLB, which assumes fixed parameters and independent measurements, the BCRLB incorporates prior information and evolves as new measurements become available [44].

The Bayesian Fisher information matrix (BFIM) for the PD position tracking process, denoted as $\mathbf{J}_P^{(B)}[t]$, is computed by combining the contributions from the measurement model in (18) and the prior information from dynamic motion in (5) can be calculated recursively as:

$$\mathbf{J}_P^{(B)}[t] = \underbrace{\mathbf{J}_P^{(O)}[t]}_{\text{Measurements}} + \underbrace{\mathbf{J}_P^{(I)}[t]}_{\text{Priori Information}}. \quad (23)$$

And after extract $\mathbf{J}_P^{(I)}$ Priori part it can be expressed as:

$$\mathbf{J}_P^{(B)}[t] = \underbrace{\mathbf{J}_P^{(O)}[t]}_{\text{Measurements}} + \underbrace{\mathbf{G}_{22}[t] - \mathbf{G}_{21}[t] \left[\mathbf{J}_P^{(B)}[t-1] + \mathbf{G}_{11}[t] \right]^{-1} \mathbf{G}_{12}[t]}_{\text{Priori-Information}}, \quad (24)$$

$$\mathbf{J}_P^{(O)}[t] = \begin{bmatrix} \sum_{i=1}^M \mathbf{J}_d^{(O)}[t]_{(i,i)} \frac{(x - x_i)^2}{(x - x_i)^2 + (y - y_i)^2} & \sum_{i=1}^M \mathbf{J}_d^{(O)}[t]_{(i,i)} \frac{(x - x_i)(y - y_i)}{(x - x_i)^2 + (y - y_i)^2} \\ \sum_{i=1}^M \mathbf{J}_d^{(O)}[t]_{(i,i)} \frac{(x - x_i)(y - y_i)}{(x - x_i)^2 + (y - y_i)^2} & \sum_{i=1}^M \mathbf{J}_d^{(O)}[t]_{(i,i)} \frac{(y - y_i)^2}{(x - x_i)^2 + (y - y_i)^2} \end{bmatrix}. \quad (22)$$

where $\mathbf{J}_P^{(1)}[t]$, quantifies the amount of useful information available for the tracking process based on the PD transition model. The $\mathbf{G}[t]$ matrices are related to $\mathbf{J}_P^{(1)}[t]$ and are formally defined as:

$$\mathbf{G}_{ij}[t] = \mathbb{E} \left(-\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t+i-2] \partial \mathbf{P}[t+j-2]^T} \right), \quad (25)$$

where $i, j = 1, 2$. Here, $p(\mathbf{P}[t] | \mathbf{P}[t-1])$ is the conditional probability of $\mathbf{P}[t]$ given $\mathbf{P}[t-1]$. after detailed mathematical calculations, the quantities $\mathbf{G}_{11}[t]$, $\mathbf{G}_{12}[t]$, $\mathbf{G}_{21}[t]$, and $\mathbf{G}_{22}[t]$ can be derived and obtained as described in Appendix section. Hence $\mathbf{G}[t]$ are expressed as [45]:

$$\mathbf{G}_{11}[t] \in \mathbb{R}^{2 \times 2} = \mathbf{Q}^{-1}, \quad (26a)$$

$$\mathbf{G}_{12}[t] \in \mathbb{R}^{2 \times 2} = -\mathbf{Q}^{-1}, \quad (26b)$$

$$\mathbf{G}_{21}[t] \in \mathbb{R}^{2 \times 2} = -\mathbf{Q}^{-1}, \quad (26c)$$

$$\mathbf{G}_{22}[t] \in \mathbb{R}^{2 \times 2} = \mathbf{Q}^{-1}. \quad (26d)$$

Finally, the BCRLB for the position at the t -th sample is determined by calculating the inverse of the BFIM as defined in (23). Consequently, the Position Error Bound (PEB) which serves as a benchmark for the variance of the position estimation error at the t -th sample, is computed as follows:

$$\text{PEB}[t] = \sqrt{\left[\mathbf{J}_P^{(B)}[t] \right]_{(1,1)}^{-1} + \left[\mathbf{J}_P^{(B)}[t] \right]_{(2,2)}^{-1}}. \quad (27)$$

Remark 8: As indicated by (24), the BFIM is affected by the process noise. When the process noise increases, the BFIM decreases, which leads to an increase in the BCRLB. It is worth mentioning that as the measurements become less noisy, the contribution of the prior information decreases, and vice versa.

IV. POSITION ESTIMATION AND TRACKING

This section focuses on estimating and tracking the PD target's position in a MISO system featuring M transmitters light sources and a single PD receiver. It explores how the measured distances from the light sources contribute to estimating the PD's location using various methods. Furthermore, this section describes how the EKF can track and estimate the PD location under varying noise conditions.

A. Distance Estimation

To determine the PD position, it is first necessary to estimate the distances between light sources and PD and we consider these distances as our measurements. At the t -th time sample, the true distances vector is defined as:

$$\mathbf{d}[t] = [d_1[t], d_2[t], \dots, d_M[t]]^T. \quad (28)$$

To obtain the estimated distance from the $\mathbf{r}[t]$, this work adopts the MLE framework to estimate the distances, explicitly considering the impact of noise and the effect of signal-dependent noises during the estimation process. The estimation is obtained by maximizing the following cost function [46]:

$$\hat{\mathbf{d}}[t] = \arg \max_{\mathbf{d}[t]} \ln f(\mathbf{r}[t] | \mathbf{d}[t]), \quad (29)$$

where the log-likelihood function from (12) is expressed as:

$$\begin{aligned} \ln f(\mathbf{r}[t] | \mathbf{d}[t]) &= -\frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_s[t]| \\ &\quad - \frac{1}{2} (\mathbf{r}[t] - \boldsymbol{\mu}[t])^\top \boldsymbol{\Sigma}_s^{-1}[t] (\mathbf{r}[t] - \boldsymbol{\mu}[t]). \end{aligned} \quad (30)$$

To estimate each an individual distance $d_i[t]$, the log-likelihood function is maximized by solving:

$$\frac{\partial \ln f(\mathbf{r}[t] | \mathbf{d}[t])}{\partial d_i[t]} = 0. \quad (31)$$

Due to the complexity of the resulting equations, finding a closed-form solution for $\hat{d}_i[t]$ is typically infeasible. Instead, numerical methods and computational tools, such as MATLAB, are employed to compute the solution efficiently.

B. Measurements Model

After estimating the distances we can assume the observation vector, $\hat{\mathbf{\Gamma}}[t]$, consists of the estimated distances:

$$\hat{\mathbf{\Gamma}}[t] = [\hat{d}_1[t], \dots, \hat{d}_i[t], \dots, \hat{d}_M[t]]^\top \quad (32)$$

The estimated distance vector $\hat{\mathbf{\Gamma}}[t]$ inherently contains noise and serves as an approximation of the true distance-vector $\mathbf{\Gamma}[t]$. The relationship between the estimated and true values is described as:

$$\hat{\mathbf{\Gamma}}[t] = \mathbf{\Gamma}[t] + \mathbf{w}[t], \quad (33)$$

where $\mathbf{w}[t] \in \mathbb{R}^{M \times 1}$ is the measurement noise vector at the t -th sample, with a covariance matrix $\mathbf{C}[t]$, which quantifies the uncertainty in the estimation. Under a Maximum Likelihood Estimation (MLE) framework, the variance of the estimation error for $d_i[t]$ typically approaches the CRLB. Thus, the CRLB serves as a valuable benchmark for evaluating how accurately $d_i[t]$ can be used to estimate position [40], [47]. Therefore, the covariance matrix $\mathbf{C}[t]$ can be expressed as:

$$\mathbf{C}[t] = \left(\mathbf{J}_d^{(O)}[t] \right)^{-1}, \mathbf{C}[t] \in \mathbb{R}^{M \times M}, \quad (34)$$

where $\mathbf{J}_d^{(O)}[t]$ is the DFIM, as derived in (17). Therefore, $\hat{\mathbf{\Gamma}}[t]$ can be expressed as:

$$\hat{\mathbf{\Gamma}}[t] \sim \mathcal{CN}(\mathbf{\Gamma}[t], \mathbf{C}[t]). \quad (35)$$

The estimated values, $\hat{\mathbf{\Gamma}}[t]$, are dependent on the position state vector $\mathbf{P}[t]$ through a functional relationship:

$$\hat{\mathbf{\Gamma}}[t] = \mathbf{f}(\mathbf{P}[t]) + \mathbf{w}[t], \quad (36)$$

where $\mathbf{f}(\mathbf{P}[t])$ represents the mapping function that relates the position vector $\mathbf{P}[t]$ to the true distances $\mathbf{\Gamma}[t]$ [48]. This model highlights how the underlying signal parameters influence the measurements. The estimation accuracy depends on the signal quality and the statistical properties of the noise vector $\mathbf{w}[t]$, both of which directly impact the subsequent tracking Processes [40].

C. Extended Kalman Filter (EKF)

Kalman filter is a widely used tool for sequentially estimating the states of dynamic systems under uncertainty. It is an optimal minimum mean square error (MMSE) estimator when both the system dynamics and observation models are linear, and the noise is Gaussian. However, many real-world systems exhibit nonlinearities in the relationship between the state vector and the observations, making the standard Kalman filter inapplicable [49]. To address this, the EKF extends the Kalman filter framework to nonlinear systems by linearizing the nonlinear models around the current state estimate using a first-order Taylor expansion [50]. This linearization enables the EKF to approximate the estimation process for nonlinear systems, albeit with some loss in optimality.

In the tracking problem, the location vector $\mathbf{P}[t]$, representing the position of the system, evolves dynamically over time and is subject to process noise. The observations $\hat{\mathbf{F}}[t]$ depend nonlinearly on $\mathbf{P}[t]$, where $f(\mathbf{P}[t]) \in \mathbb{R}^{M \times 1}$. The EKF addresses this by approximating the nonlinear observation function $f(\mathbf{P}[t])$ with its first-order Taylor expansion around the current estimate $\hat{\mathbf{P}}[t|t-1]$. The Jacobian matrix of this linearized model, denoted by $\mathbf{F}[t]$, is computed as [44] and [49]:

$$\mathbf{F}[t] \in \mathbb{R}^{M \times 2} = \left. \frac{\partial f(\mathbf{P}[t])}{\partial \mathbf{P}[t]} \right|_{\mathbf{P}[t]=\hat{\mathbf{P}}[t|t-1]}. \quad (37)$$

The function $f(\mathbf{P}[t])$ is defined as:

$$f(\mathbf{P}[t]) = \begin{bmatrix} \sqrt{(x[t] - x_1)^2 + (y[t] - y_1)^2} \\ \vdots \\ \sqrt{(x[t] - x_i)^2 + (y[t] - y_i)^2} \\ \vdots \\ \sqrt{(x[t] - x_M)^2 + (y[t] - y_M)^2} \end{bmatrix}. \quad (38)$$

The derivative with respect to $x[t]$ is:

$$\frac{\partial f_i(\mathbf{P}[t])}{\partial x[t]} = \frac{x[t] - x_i}{\sqrt{(x[t] - x_i)^2 + (y[t] - y_i)^2}}. \quad (39)$$

Similarly, the derivative with respect to $y[t]$ is:

$$\frac{\partial f_i(\mathbf{P}[t])}{\partial y[t]} = \frac{y[t] - y_i}{\sqrt{(x[t] - x_i)^2 + (y[t] - y_i)^2}}. \quad (40)$$

Consequently, the linearized version of the measurements equation in (36) can be given as:

$$\begin{aligned} \hat{\mathbf{F}}[t] &= \mathbf{F}[t]\mathbf{P}[t] + \underbrace{\mathbf{w}[t]}_{\text{Measurements error}} \\ &\quad + \underbrace{\left(f(\hat{\mathbf{P}}[t|t-1]) - \mathbf{F}[t]\hat{\mathbf{P}}[t|t-1] \right)}_{\text{linearization error}}. \end{aligned} \quad (41)$$

The EKF operates recursively through prediction and correction steps to refine the state estimate $\hat{\mathbf{P}}[t|t]$. In the prediction step, the state is forecasted using the previous estimate as [41]:

$$\hat{\mathbf{P}}[t|t-1] = \hat{\mathbf{P}}[t-1|t-1] + \mathbf{q}[t|t-1], \quad (42)$$

and the error covariance matrix is updated as:

$$\mathbf{E}[t|t-1] = \mathbf{E}[t-1|t-1] + \mathbf{Q}, \quad (43)$$

where $\mathbf{Q}$ is the process noise covariance matrix. In the correction step, the Kalman gain $\mathbf{K}[t]$ is computed as:

$$\mathbf{K}[t] = \mathbf{E}[t|t-1]\mathbf{F}^\top[t] \left(\mathbf{C}[t] + \mathbf{F}[t]\mathbf{E}[t|t-1]\mathbf{F}^\top[t] \right)^{-1}. \quad (44)$$

Finally, the updated state estimate is:

$$\hat{\mathbf{P}}[t|t] = \hat{\mathbf{P}}[t|t-1] + \mathbf{K}[t] \left(\hat{\mathbf{F}}[t] - f(\hat{\mathbf{P}}[t|t-1]) \right), \quad (45)$$

and the error covariance is revised as:

$$\mathbf{E}[t|t] = (\mathbf{I} - \mathbf{K}[t]\mathbf{F}[t])\mathbf{E}[t|t-1]. \quad (46)$$

In this context, the tracking process begins by initializing Equations (42) and (43) with $\hat{\mathbf{P}}[-1|-1] = \mathbf{P}_0$ and $\mathbf{E}[-1|-1] = \mathbf{Q}_0$. Here, (42) corresponds to the prediction step, formulated based on the transition model given in (5), while (43) calculates the minimum prediction MSE matrix, $\mathbf{E} \in \mathbb{R}^{2 \times 2}$. Additionally, equation (44) determines the Kalman gain matrix $\mathbf{K}[t] \in \mathbb{R}^{2 \times M}$ based on the measurement error matrix $\mathbf{C}[t]$, which quantifies the uncertainties in the measurement process and is defined in (34). Subsequently, the correction step in (45) refines the prediction $\hat{\mathbf{P}}[t|t-1]$ from (42) by incorporating the current measurements, $\hat{\mathbf{F}}[t]$. This correction step aims to produce a more accurate estimate of the position vector, $\hat{\mathbf{P}}[t|t]$. The output, $\hat{\mathbf{P}}[t|t]$, represents the tracker's result for the t -th sample, which depends on all the information from the initial sample up to the t -th sample. The minimum MSE matrix for the updated state is given in (46). The updated $\hat{\mathbf{P}}[t|t]$ and $\mathbf{E}[t|t]$ are then re-fed as prior values for the next prediction step in (42), (43) [6], [51].

V. SIMULATION RESULTS AND DISCUSSION

Simulations were conducted using MATLAB to evaluate the performance of the MISO-VLP system under the joint effects of SDSN and RIN. Approximately 10^6 Monte Carlo iterations were performed to ensure statistical reliability. In each simulation run, six light sources were positioned within a $4\text{ m} \times 4\text{ m}$ room. The placement was constrained to guarantee that, at every time step, the receiver remained within the FOV of at least three light sources. This ensures robust signal reception and enables accurate trilateration.

All light sources were configured with identical physical and optical parameters, including a FOV angle of $\phi_{\text{FOV}} = 70^\circ$, vertical separation $L = 3\text{ m}$, active detection area $A_{\text{pd}} = 1\text{ cm}^2$, and optical filter gain $T_s(\theta_{\text{rx}}) = 1$. The noise variances were defined as $\sigma_x^2 = 10^{-3}$, $\sigma_y^2 = 10^{-3}$, $\sigma_R^2 = 10^{-3}$, $\sigma_{\text{ds}}^2 = 10^{-1}$, and $\sigma_n^2 = 10^{-1}$. Finally, without losing the generality, and to illustrate a representative scenario, six fixed light sources configuration was also analyzed with coordinates located at $[1, 1, 3]$, $[2, 1, 3]$, $[3, 1, 3]$, $[1, 3, 3]$, $[2, 3, 3]$, $[3, 3, 3]$. However, we would like to emphasize that our analysis is valid for any light source positions. The PD was initialized at $\mathbf{P}_0 = [2, 2, 0]$ meters in a two-dimensional plane and moved with a velocity limited to 1 m/s along both x -axis and y -axis. Pilot signals were transmitted at regular intervals of $\Delta = 0.01$ seconds and $\mathbf{a} = [0, 0]$. These are the default parameters unless stated otherwise.

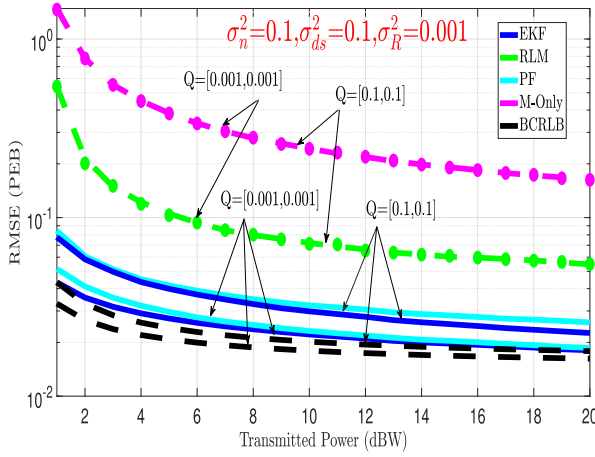


Fig. 3. RMSE comparison for the EKF, *M*-Only, RLM, PF, and the PEB from BCRLB as in (27) under $\mathbf{Q} = [0.001, 0.001]$ and $\mathbf{Q} = [0.01, 0.01]$, with noise variances $\sigma_n^2 = 0.1$, $\sigma_{ds}^2 = 0.1$, and $\sigma_R^2 = 0.001$.

In this work, we propose the use of the EKF for real-time tracking of a moving PD in VLP systems. The EKF integrates a dynamic motion model with nonlinear distance-based measurements to recursively estimate the PD's position in the presence of SDSN and RIN. In Fig. 3, we present a comparative RMSE analysis of four estimation techniques, including our proposed EKF-based localization method, to evaluate their performance in a VLP environment. The Measurement-Only (*M*-only) method estimates the PD position using geometric relationships derived from noisy distance measurements, typically solved through trilateration or numerical techniques, as described in detail in [52] and [53] and the References therein. The RLM algorithm applies a nonlinear least-squares optimization for real-time tracking, but lacks explicit modeling of system dynamics and process noise, limiting its robustness in dynamic scenarios and described in detail in [54], [55], and [56] and the references therein. The PF uses a sampling-based Bayesian approach that is robust to nonlinearities and non-Gaussian noise but is computationally intensive, especially as the number of particles increases, and is described in detail in [37] and [57], and the references therein. Our proposed EKF method combines both motion and measurement models in a recursive framework, offering a favorable balance between accuracy and computational complexity.

Moreover, the results in Fig. 3 show that the RMSE curves of the EKF and PF estimators (with the number of particles set to 100) are closely matched and consistently outperform those of the RLM and *M*-only methods. Additionally, when varying the process noise covariance $\mathbf{Q}$, the RMSE of both *M*-only and RLM (with a damping factor set to 0.01) remains unchanged, as expected, since these methods do not model motion dynamics and are therefore insensitive to process noise. In this figure, the PD moves with linear velocity with the acceleration vector is defined as $\mathbf{a} = [0.1, 0.1] \text{ m/s}^2$.

In contrast, the RMSE for EKF and PF increases with larger $\mathbf{Q}$, along with a corresponding rise in the PEB calculated from BCRLB as in (27). This behavior highlights their sensitivity to process uncertainty and reflects their reliance on accurate dynamic modeling. These findings align with the theoretical

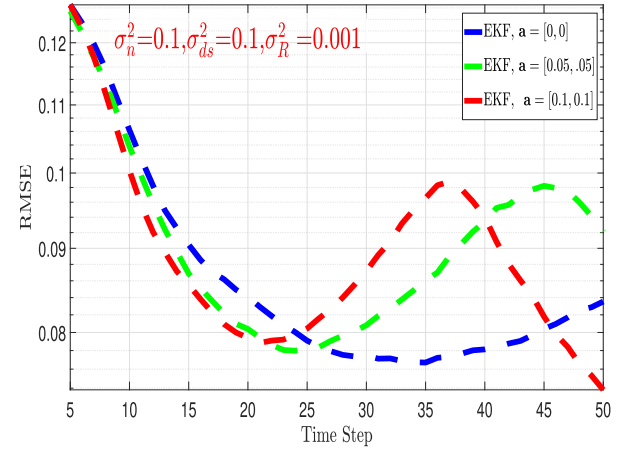


Fig. 4. Compare RMSE over time for EKF at different values of acceleration under $\mathbf{Q} = [0.001, 0.001]$ and with noise variances $\sigma_n^2 = 0.1$, $\sigma_{ds}^2 = 0.1$, and $\sigma_R^2 = 0.001$.

prediction in (24), where a higher process noise variance reduces the BFIM, resulting in an increased BCRLB and thereby validating the simulation results.

In Fig. 4, we assume that the PD moves with a velocity that changes linearly over time at acceleration, corresponding to different acceleration values, $\mathbf{a} = [0, 0]$, $\mathbf{a} = [0.05, 0.05]$ and $\mathbf{a} = [0.1, 0.1]$. The transmitted optical power is fixed at $p = 10$ Watts. As observed from the figure, the RMSE initially decreases for the EKF estimator across all acceleration levels. This trend suggests that the relative distance between the PD and the light sources decreases during the early stages of motion, indicating that the PD is generally moving toward the light source cluster. However, after a certain point, the RMSE begins to increase. This behavior occurs because, although the PD may be approaching some light sources, it is simultaneously moving farther away from others, thereby increasing the total relative distance from the entire set of light sources. In multi-source localization scenarios, proximity to a single light source does not necessarily guarantee improved accuracy, as estimation performance depends on the relative geometry and cumulative distances to all transmitters. Furthermore, the figure shows that higher acceleration leads to earlier shifts in the RMSE trend compared to lower acceleration. This observation highlights the role of acceleration in modulating the motion dynamics of the PD and its impact on localization performance.

Fig. 5 illustrates the RMSE performance of the EKF for a PD localizing under three light sources positioned at coordinates $[1, 1, 3]$, $[2, 1, 3]$ and $[3, 1, 3]$. The PD is initialized at $[2, 2, 0]$ and moves linearly along the y -axis under two different acceleration magnitudes: $\mathbf{a} = [0, 0.05] \text{ m/s}^2$ and $\mathbf{a} = [0, 0.1] \text{ m/s}^2$. The motion is examined in two cases: the PD moving toward the light sources and moving away from them. In both cases, the RMSE behavior is strongly influenced by the distance between the PD and the light sources. When the PD moves toward the light sources, the RMSE is lower than when it moves away from them. This is due to the inverse dependence of signal strength on distance, which directly affects measurement quality and aligns with the (22) and

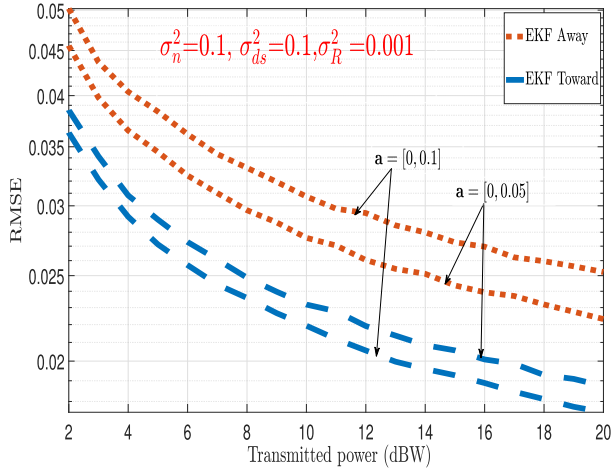


Fig. 5. Compare RMSE of EKF and at different values of acceleration under $\mathbf{Q} = [0.001, 0.001]$ and with noise variances $\sigma_n^2 = 0.1$, $\sigma_{ds}^2 = 0.1$, and $\sigma_R^2 = 0.001$ in two cases toward light sources and away from them.

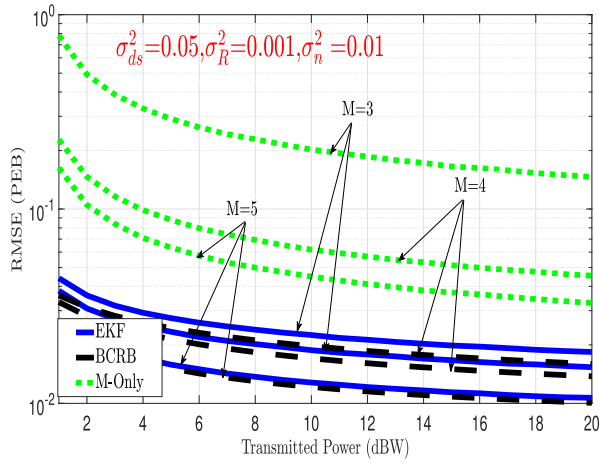


Fig. 6. RMSE comparison for the EKF and the PEB from BCRLB as in (27) for varying numbers of light sources ($M = 3, 4, 5$), with noise variances $\sigma_n^2 = 0.01$, $\sigma_{ds}^2 = 0.05$, and $\sigma_R^2 = 0.001$.

Remark 1. Furthermore, the effect of acceleration magnitude is notable. In the scenario where the PD accelerates toward the light sources, the higher magnitude $\mathbf{a} = [0, 0.1]$, m/s^2 results in a faster reduction in RMSE. Conversely, when the PD accelerates away from the light sources, the same magnitude leads to a more rapid increase in RMSE compared to the lower acceleration $\mathbf{a} = [0, 0.05]$ m/s^2 .

Fig. 6 illustrates the impact of the number of light sources on the RMSE of location estimation where $\sigma_n^2 = 0.01$, $\sigma_{ds}^2 = 0.05$ and $\sigma_R^2 = 0.001$. The results demonstrate that increasing the number of light sources improves the accuracy of the location estimation, reducing the RMSE in both the M-only approach and the EKF. Additionally, the EKF performance approaches PEB calculated from the BCRLB as the number of light sources increases. This improvement is consistent with the theoretical relationship described in (22), which highlights the dependency of the PFIM on the number of light sources, M . As M increases, the value of elements of the PFIM, $\mathbf{J}_P^{(O)}[t]$, increases, leading to a reduction in the PEB. This result confirms that incorporating more light sources

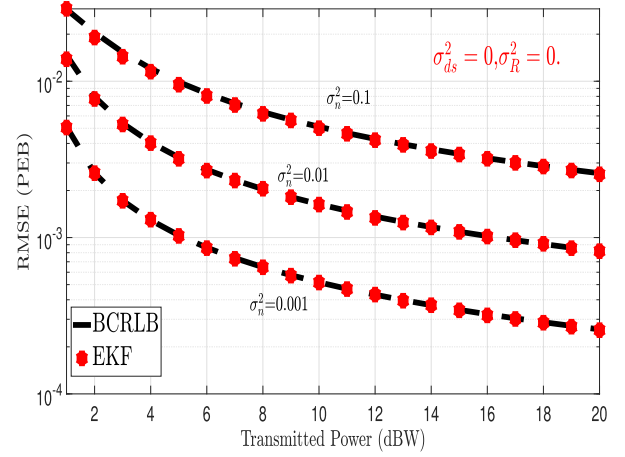


Fig. 7. RMSE comparison for the EKF and the PEB from BCRLB as in (27) under varying thermal noise variances, with $\sigma_{ds}^2 = 0$ and $\sigma_R^2 = 0$.

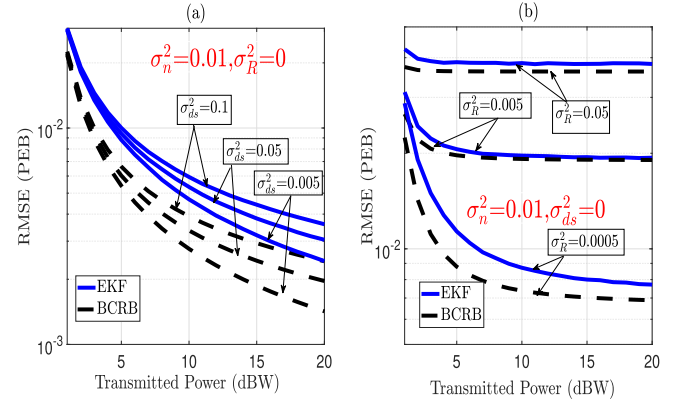


Fig. 8. RMSE comparison for the EKF and the PEB from the BCRLB as in (27) under varying RIN and SDSN noise variances, with $\sigma_n^2 = 0.01$.

enhances the precision of position estimation by providing greater spatial diversity and improving system observability.

Fig. 7 illustrates the performance of the EKF in location estimation under the scenario where σ_n^2 is non-zero, $\sigma_{ds}^2 = 0$, $\sigma_R^2 = 0$, and $\mathbf{v}(t) = [0, 0]$. In this case, the only source of error from measurements is thermal noise with variance σ_n^2 . The results demonstrate that, in the absence of SDSN and RIN, the EKF acts as the optimal estimator, with its RMSE of EKF closely aligning with the PEB derived for the BCRLB. Furthermore, the figure highlights that as the value of σ_n^2 increases, the RMSE also rises, which is consistent with the theoretical results presented in (17) and Remark 2. This confirms the expected dependency of estimation accuracy on noise levels. The performance comparison of the EKF and the PEB from the BCRLB is presented in Fig. 8. In Fig. 8 (a), we consider the case where $\sigma_R^2 = 0$, $\sigma_n^2 = 0.01$, and σ_{ds}^2 varies. It can be observed that the presence of SDSN prevents the RMSE of EKF and PEB from the BCRLB from becoming identical, unlike the results shown in Fig. 7. As the value of σ_{ds}^2 increases, both the RMSE and the PEB from the BCRLB increase, leading to a degradation in estimation performance.

In Fig. 8 (b), the scenario is reversed: $\sigma_{ds}^2 = 0$, while σ_R^2 varies, with σ_n^2 set to 0.01, similar to Fig 8 (a). The RMSE is larger when σ_R^2 is present compared to when σ_{ds}^2

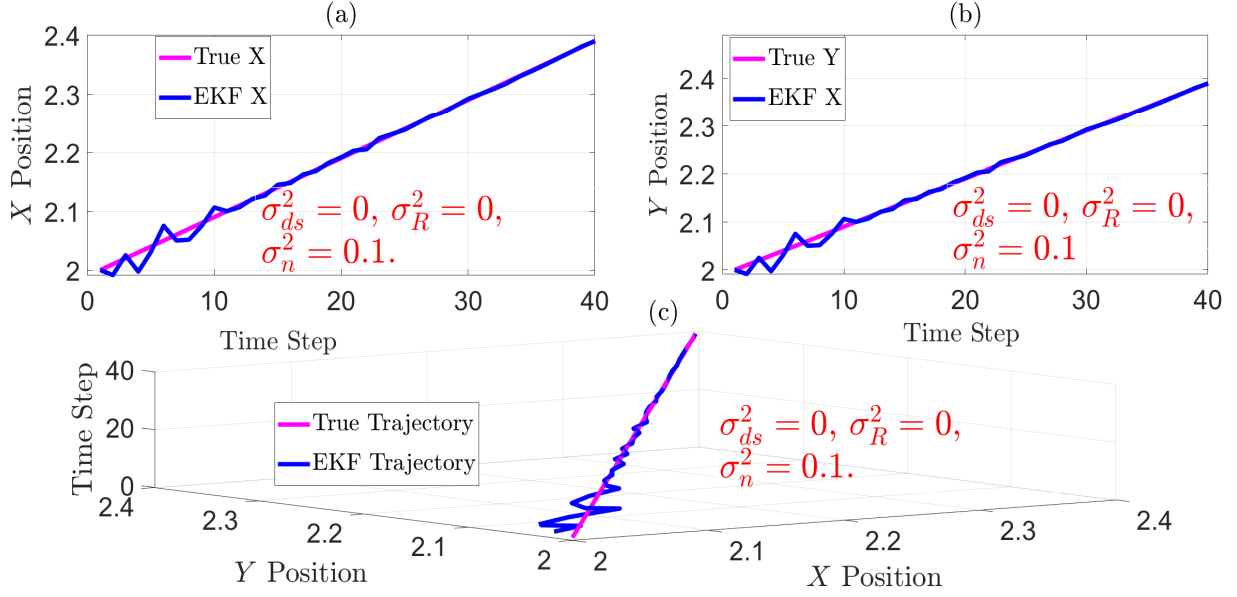


Fig. 9. Position tracking comparison between the EKF trajectory and the true trajectory in the ideal case with $\sigma_R^2 = \sigma_{ds}^2 = 0$ and $\sigma_n^2 = 0.1$.

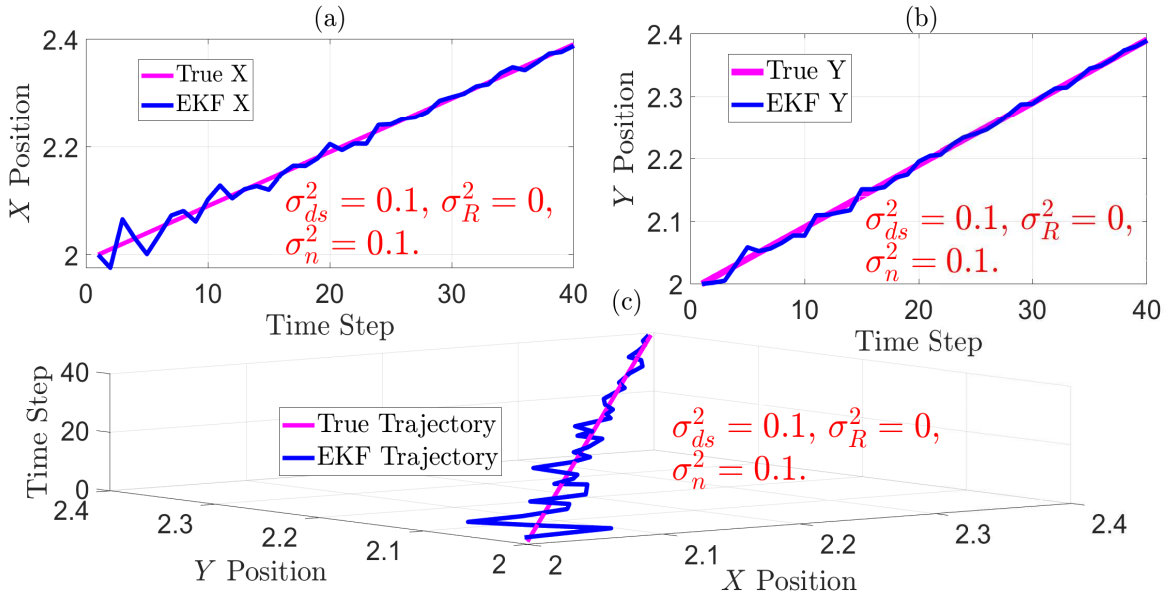


Fig. 10. Position tracking comparison between the EKF trajectory and the true trajectory in the case of no RIN noise and the presence of SDSN noise with $\sigma_{ds}^2 = 0.1$ and $\sigma_n^2 = 0.1$.

is the dominating noise. Additionally, the RMSE and PEB from BCRLB saturate at specific values, which does not occur in the case of SDSN, as shown in Fig. 8 (a). This behavior is consistent with the theoretical results derived in (17) and Remark 6. As shown in (17), σ_s^2 includes σ_R^2 multiplied by p^2 and σ_{ds}^2 multiplied by p . Consequently, increasing the RIN worsens the estimation accuracy more significantly than SDSN. This is further demonstrated in Fig. 8, where even though the values of σ_{ds}^2 are larger than σ_R^2 , the RMSE does not reach saturation in the case of SDSN.

Fig. 9 shows the trajectory of the PD target location along the x - and y -axes in (a) and (b) sub-figures in sequence and in the 2D plane in the (c) sub-figure for $\sigma_{ds}^2 = 0$, $\sigma_n^2 = 0.1$, and $\sigma_R^2 = 0$. As observed with $\sigma_R^2 = 0$ and $\sigma_{ds}^2 = 0$, the EKF accurately tracks the moving target. However, in the presence of thermal noise ($\sigma_n^2 > 0$), the EKF learns to compensate for the thermal noise effect over time, as it is independent of the transmitted power. This aligns with the findings in Fig. 7.

In Fig. 10, we analyze the scenario where $\sigma_{ds}^2 = 0.1$, $\sigma_n^2 = 0.1$, and $\sigma_R^2 = 0$. The results indicate that the presence of

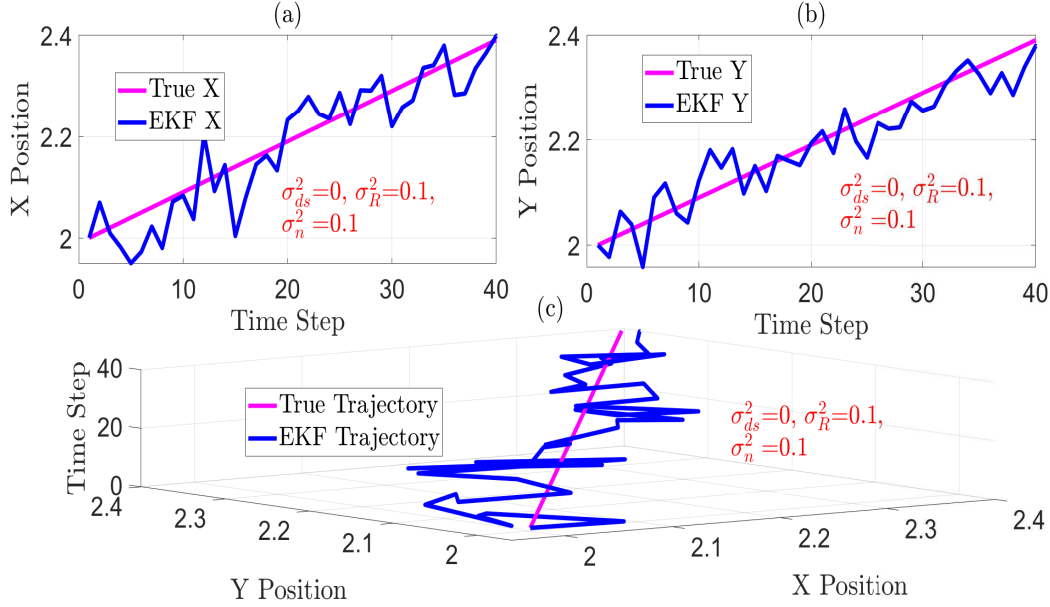


Fig. 11. Position tracking comparison between the EKF trajectory and the true trajectory in the case of no SDSN noise and the presence of RIN noise with $\sigma_R^2 = 0.1$ and thermal noise with $\sigma_n^2 = 0.1$.

SDSN noise significantly degrades the tracking accuracy of the EKF, and we can see that in x-y axes tracking in subfigures (a), (b), and 2D Plane in subfigure (c). This degradation stems from imperfections in the transmitter and receiver materials, which introduce signal-dependent noise that adversely affects the efficiency of the tracking process. As described in (41), the measurement noise variance is directly influenced by the variance of SDSN, leading to reduced estimation performance.

In Fig. 11, we analyze the impact of RIN noise on tracking accuracy. The results demonstrate that RIN noise significantly degrades the precision of target tracking. As explained in (41), the variance of RIN noise increases with the quadratic transmitted power, severely impairing the EKF's ability to track the target's position accurately. This emphasizes the EKF's sensitivity to power-dependent noise sources such as RIN and SDSN, further highlighting the challenges these noise types pose in maintaining tracking efficiency. We can see the RIN effect in the x -axis in Fig. 11 (a), in the y -axis in Fig. 11 (b), and in the $x-y$ plane in Fig. 11 (c).

VI. CONCLUSION

In this paper, we investigated position estimation and tracking in indoor VLC systems using a MISO-VLC setup with multiple light sources as transmitters. An EKF was applied for real-time position tracking, and its performance was compared against several alternative techniques. These included a measurement-only approach based on solving geometric equations from noisy distances, the RLM method, and a PF. The results demonstrated that the EKF achieves significantly lower RMSE in estimated positions. While PF offers similar accuracy, it incurs higher computational complexity, and RLM and measurement-only only show less robustness due to their sensitivity to nonlinearity and lack of state propagation.

Additionally, we evaluated the impact of increasing the number of light sources on localization accuracy. Our findings indicate that as the number of light sources increases and the target remains within their effective coverage, the spatial diversity improves, leading to enhanced estimation precision. We further investigated the effects of signal-dependent noise sources, specifically RIN and SDSN. The results showed that higher levels of RIN degrade both tracking and estimation performance more severely than SDSN. To establish a theoretical performance benchmark, we derived a closed-form BCRLB. The analysis revealed that the BCRLB is influenced by SDSN, RIN, and process noise, validating the EKF's effectiveness against this theoretical bound and confirming its suitability for dynamic VLC-based localization systems. It is worth noting that the current work assumes line-of-sight (LoS) propagation. In practice, NLoS components may arise, e.g., due to reflections, potentially impacting the estimator's accuracy. Therefore, extending the proposed VLP and tracking framework to account for NLoS conditions in conjunction with signal-dependent noise represents an important direction for future research.

APPENDIX

This appendix provides the detailed derivation of the prior information FIM, denoted as $\mathbf{J}_P^{(I)}$. The prior information about the PD transition is modeled statistically as Gaussian in (5). The log function of the Gaussian PDF can be expressed as:

$$\begin{aligned} \log p(\mathbf{P}[t] | \mathbf{P}[t-1]) = & A - \frac{1}{2} \log |\mathbf{Q}| \\ & - \frac{1}{2} (\mathbf{P}[t] - (\mathbf{P}[t-1] + \mathbf{q}[t]))^T \mathbf{Q}^{-1} \\ & \times (\mathbf{P}[t] - (\mathbf{P}[t-1] + \mathbf{q}[t])), \quad (47) \end{aligned}$$

where A is a constant, and $\mathbf{Q}$ is a matrix representing process noise. The first derivatives of the log-likelihood with respect to $\mathbf{P}[t]$ and $\mathbf{P}[t-1]$ are:

$$\begin{aligned} \frac{\partial \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t]} \\ = -(\mathbf{Q})^{-1} (\mathbf{P}[t] - (\mathbf{P}[t-1] + \mathbf{q}[t])), \end{aligned} \quad (48)$$

and

$$\begin{aligned} \frac{\partial \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t-1]} \\ = (\mathbf{Q})^{-1} (\mathbf{P}[t] - (\mathbf{P}[t-1] + \mathbf{q}[t])). \end{aligned} \quad (49)$$

The second derivatives of the log-likelihood are:

$$\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t] \partial \mathbf{P}[t]^T} = -(\mathbf{Q})^{-1}, \quad (50)$$

$$\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t-1] \partial \mathbf{P}[t-1]^T} = -(\mathbf{Q})^{-1}, \quad (51)$$

$$\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t] \partial \mathbf{P}[t-1]^T} = (\mathbf{Q})^{-1}, \quad (52)$$

$$\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t-1] \partial \mathbf{P}[t]^T} = (\mathbf{Q})^{-1}. \quad (53)$$

Using these derivatives, the components of the Fisher information matrix ($\mathbf{G}[t]$) are defined as follows:

$$\mathbf{G}_{11}[t] = \mathbb{E} \left(-\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t-1] \partial \mathbf{P}[t-1]^T} \right) = (\mathbf{Q})^{-1}, \quad (54a)$$

$$\mathbf{G}_{12}[t] = \mathbb{E} \left(-\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t-1] \partial \mathbf{P}[t]^T} \right) = -(\mathbf{Q})^{-1}, \quad (54b)$$

$$\mathbf{G}_{21}[t] = \mathbb{E} \left(-\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t] \partial \mathbf{P}[t-1]^T} \right) = -(\mathbf{Q})^{-1}, \quad (54c)$$

$$\mathbf{G}_{22}[t] = \mathbb{E} \left(-\frac{\partial^2 \log p(\mathbf{P}[t] | \mathbf{P}[t-1])}{\partial \mathbf{P}[t] \partial \mathbf{P}[t]^T} \right) = (\mathbf{Q})^{-1}. \quad (54d)$$

which concludes the proof.

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